BrickPlan 1.2: Auditable Visual Diagnostics of Assembly-Order Errors and Recovery

Executable benchmark research release

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Abstract

An assembly action can be physically admissible yet make later work more expensive. BrickPlan measures this distinction using certified, finite assembly transitions and exact completion costs. This revision corrects trajectory-level measurement, replaces shortcut-prone continuation probes, and adds matched raster-image and structured observation tracks. The release contains 3 structures from two source groups, 15 episodes and 87 probe responses. All 342,704 reachable states and 2,843,080 directed transitions retain the source geometry and fixed-path qualifications of the preceding release. Thirty-six continuation pairs preserve initial observations, first and last actions, and action multisets while changing interior order. A first–last shortcut can no longer solve them. Self-induced feasibility losses now have explicit detection, restoration and censoring records. The revision also exposes a substantive limitation: a privileged static relation rule solves all 15 episodes. Consequently, this is an auditable diagnostic release, not evidence of search hardness or demonstrated VLM shortsightedness. No model or human performance has been measured.

1 Motivation and scope

End-to-end assembly success conflates perception, immediate legality, future access, and recovery. A closure can be installed correctly while blocking a required interior insertion. Such an action is locally legal; rejecting it using a goal-aware simulator would erase the error the benchmark seeks to measure. BrickPlan therefore separates goal-independent execution from evaluator-only future-feasibility labels.

Interactive reconstruction and assembly are established tasks, including LTRON [1], InstructioNet [2], LEGO-Puzzles [3], PhyBlock [4], and Brick-Composer [5]. Our contribution is the measurement and audit contract linking legal decisions to later rework. We do not claim that interaction, disassembly, or multi-step construction is new.

An adversarial audit of version 1.1 found that a static blocker-closure rule solved every episode optimally and a first–last-action shortcut solved every continuation probe. It also identified a hard-coded two-structure census in the three-structure report, checkpoint-only dead-end measurement, and a structured payload unsuitable for evaluating visual perception. Version 1.2 repairs these concrete issues without overwriting frozen 1.0 or 1.1 artifacts. An increased episode length is not treated as increased reasoning depth.

Table 1: Study conditions. These are unfilled experimental slots, not results. Exact model identifiers, prompts, budgets and providers must be registered before collection.

Condition	Status
Same VLM, image+ID, stepwise	Not evaluated
Same VLM, image+ID, explicit planning	Not evaluated
Same VLM, structured control	Not evaluated
Human, image+ID protocol	Not evaluated
Human, structured control	Not evaluated

2 Experimental design before empirical results

Each condition uses the same task starts, goals, identity seeds and action budgets. Camera changes use the same inspection API for humans and agents. The preregistration manifest records participant or model identity, provider and version, prompt hash, temperature, reasoning setting, context and output limits, provider cost cap, seed, track, timeout and retry policy. The provider-neutral driver consumes public observations through an external policy adapter. No provider calls are needed to reproduce the software controls. Currency and model-token enforcement belong to that adapter; the executor enforces action, inspection and diagnosis limits.

Evaluation runs have isolated stores and a fixed observation track. Study mode permits one start per assigned episode and does not expose scores or state replay. Administrative collection occurs on the evaluator host. Policy processes must not have access to private banks, certificates, cost tables, other tracks, or past study answers. Local practice is not this isolation boundary. Roadster and kart share one source kit and must remain in the same provenance split. Repeated seeds are repeated runs,

not additional structures. Two provenance groups cannot support strong population-level inference.

3 Certified workcell and tasks

3.1 Physical abstraction

Each site is an original LDraw part instance at its original source pose. A connected fixed subassembly is explicitly clamped. A mutable part may be attached or detached along a specified straight translation if the path is clear and the resulting assembly stays connected to the fixed base through qualified stud connections. Connections are abstract rigid holds. No artificial spacers, pose shifts, floating installed parts, or visually substituted box meshes are introduced.

Version 1.2 reuses the geometry and path evidence frozen in 1.1. Closed collision proxies are inset by 0.25 LDU per surface, with a 0.002 LDU guard. Final-pose pairs must be clear; moving–obstacle pairs cover the full declared translation. Interval clearance uses the 1-Lipschitz bound for surface distance under translation; intersections witness obstruction. Unknown geometry is inadmissible. These are numerical certificates within the declared proxy/path domain, not physical experiments or free-motion guarantees. The earlier source-verified repair of one missing 3022 proxy cap face remains documented in the frozen provenance; display triangles are unchanged.

Let s be mutable occupancy and g the exact target occupancy. Local legality uses only s, path blockers and connectivity, never g or future costs. Every legal transition has a verified inverse. Goal-rooted breadth-first search provides $C^*(s, g)$, checked against the complete reachable component through Bellman inequalities and decreasing-successor witnesses. For $\Delta = |g| - |s|$, minimum detach count is

$$D^*(s, g) = \frac{C^*(s, g) - \Delta}{2}.$$

All task starts and completions have replayable witnesses. Hashes bind source, geometry, transitions, tasks and cost tables; a trajectory also binds its full task definition, including budgets.

3.2 Episode protocols

Each structure has five episodes: full build, safe continuation checkpoint, injected repair, size-matched healthy control, and partial-target reconfiguration. Full builds contain 17, 18 and 21 mutable pieces. Injected repairs require exactly three detachments; their healthy controls match installed count and inherit the repair budget. Other action budgets are $C^* + \max(4, \lceil C^*/2 \rceil)$. Inspections have quota 32; diagnoses have quota 8. Rejected structural attempts consume one action.

Reconfiguration starts contain both pieces absent from the final goal and missing required pieces. A legal, goal-directed attachment exists that adds two completion actions. Each start also has an optimal legal alternative. These tasks exercise temporary access restoration over 24–34 actions, but they retain the known access-order mechanism. Exhaustive empty-to-state analysis found no temporary-scaffold construction in any inherited bank. A deterministic search over existing goals and 30 additional reachable partial goals per structure found no example meeting our stronger "every legal required removal is worse than an available installation" criterion. This negative search is not an impossibility proof over arbitrary source models, and support rerouting is not claimed.

3.3 Matched continuation and counterfactual probes

For each structure, 24 continuation probes form 12 pairs at lengths 4, 6 and 8. Every interior position from 2 through length minus 1 receives one geometric conflict and one matched executable control. Both plans include attachments and detachments; control steps are actual state changes. We construct a legal walk and swap two interior actions, accepting it only when authoritative replay first fails with motion obstruction or unsupported assembly at the prescribed position. Duplicate-install and empty-removal errors are excluded as the defining conflict.

The pair shares initial state, goal, randomized physical IDs, first action, last action, length and action multiset. Only interior order differs. Thus any classifier restricted to those invariant features gives the same prediction for both members and has at most 50% exact pair accuracy. The actual first–last transition shortcut obtains 36/72 correct; full transition replay obtains 72/72. This establishes resistance to that shortcut, not immunity to all shortcuts or a bound on a model's internal search.

Three inherited one-step choice probes and 12 counterfactual member responses complete the 87-response bank. Counterfactuals change the visible future goal while keeping initial state and offered actions fixed. Each source has one uniquely switching pair and one invariance pair. Evaluation uses exact conditional cost $Q^*(s, a, g) = 1 + C^*(T(s, a), g)$ and normalizes independently permuted candidate positions. These labels are defined among offered actions.

4 Matched observation tracks

The image+ID track exposes four PNG raster panels: current assembly with P/S callouts, target with S callouts, all-site reference, and a catalog of site shapes and available P inventory. Opaque P and S ID lists support action selection; their permutations remain stable within a trial. Original LDraw triangles are transformed and rasterized on the evaluator server using a z-buffer. Only pixels leave the renderer. A visible-part callout requires visible surface pixels; geometry, depth, object masks and

Table 2: Certified bank and new partial-target reconfiguration episodes. C^* and D^* refer to the reconfiguration start, not the full-build task. Hamming distance counts initially mismatched sites.

Case	Source group	Sites	States	C^*	D^*	Hamming
roadster-body	LEGO-31027	17	14,012	30	14	4
kart-deck	LEGO-31027	18	44,604	24	8	12
airframe	LEGO-31028	21	284,088	34	15	8

connectivity are not serialized into the policy payload.

The structured control receives the same images plus calibrated site matrices, types, directions, occupancy and target flags. These fields are explicitly privileged relative to image+ID, but still omit private blocker relations and costs. The visual HTTP server has no mesh, geometry-file or three.js scene-download route. It serves a canvas UI from the same raster observations the API agent receives.

The initial isometric view is free. Each subsequent inspect request selects one of seven fixed views or an exploded-reference diagram and consumes one inspection. Exploding changes only the all-site reference; current and target assemblies retain original poses. The same view changes apply to all panels. There is no free orbit during scored interaction. Probe inspection budgets persist across reloads, and answers are frozen once submitted. These are known-pose placement diagnostics with visual input, not unconstrained pose estimation or robot grasp planning.

5 Seven measurement dimensions

Dead-end choice rate. Define attach-only viability $V^+(s,g) = [s \subseteq g \wedge C^*(s,g) = |g| - |s|]$, and no-rework viability $V^0(s,g) = [C^*(s,g) = |s \triangle g|]$. The latter allows necessary deletion of extra pieces but no temporary reversal of an already matched site. Budget viability is $V_B(s,g,b) = [C^*(s,g) \le b]$. For each predicate, an online legal decision is eligible only if the predecessor is viable and both safe and unsafe legal alternatives exist. The numerator counts chosen viability losses within that denominator. Invalid actions are reported separately; they cannot enter a legal-choice denominator, but may destroy budget viability. One-step probe rates remain separate from online rates.

Lookahead depth. Score the exact first failing action, or null for a fully executable plan. Report balanced accuracy and coverage both by plan length and by the matched pair's actual conflict position. Conflict recall and valid-control false alarms accompany accuracy; missing and invalid responses remain visible. A supplied-plan distance curve is not a certified internal reasoning depth, nor does a legal continuation necessarily reach the task goal.

Plan survival curve. Freeze the first attempted-action index losing each predicate. Restoration does not erase a historical loss. Within each case, protocol, optimal length and budget stratum, report known survivors, losses and censored trials, with bounds $[A_t/N, (A_t + U_t)/N]$. Missing episodes are censored even at time zero; a successful terminated trial is absorbing. Default reports have one run per task, so curves are descriptive records rather than reliable population estimates.

Regret. For successful episodes, action regret is attempted actions minus initial C^*; detach regret is successful detach operations minus initial D^*. Failed and unfinished episodes have null regret and explicit consumed costs. Conservation gives

$$R_{\text{action}} = 2R_{\text{detach}} + N_{\text{invalid}}.$$

These are correlated views of cost, not independent evidence. Legal-action Bellman excess is separately observable to the evaluator: $1 + C^*(s_{t+1}, g) - C^*(s_t, g)$.

Recovery rate. A self-induced event begins when a legal action changes V^0 from true to false. The first diagnosis after that event is retained, even when earlier successful structural actions occurred. Its sign, attempt lag, and coverage are explicit. An event restores when V^0 becomes true again; a new later loss creates a new event. Terminal unrestored events differ from censored, unfinished events. Report restoration and joint detection/restoration over all events, plus an upper bound including censored outcomes. Injected-repair detection and completion remain a separate protocol. Absolute unreachability is not implied: the certified component is reversible. Budget loss is separately irreversible because spent actions cannot be refunded.

Replanning efficiency. For every self-induced event, retain attempts and detachments through restoration, exact cost and minimum detachments at entry, and prefix excess $k + C^*(s_{\text{restore}}, g) - C^*(s_{\text{entry}}, g)$. Report these costs alongside event coverage and restoration, not only for successful survivors. Episode-level detach count, distinct removed P IDs and failed attempts remain available. Diagnoses are also scored once per distinct visited state, preventing repeated answers at one state from inflating aggregate accuracy.

Table 3: Software controls, not VLM or human scores. Runs reuse the same 15 tasks; denominators are task-runs, not independent structures.

Software control	Runs	Success
oracle	1	15/15
static-relations	1	15/15
search-1	3	27/45
search-2	3	27/45
search-4	3	27/45
public-greedy	3	7/45
legal-random	3	0/45
blind-random	3	0/45

Counterfactual consistency. Joint correct switching requires correct offered choices in both changed-goal members. Correct invariance requires correctness in both control members. Switching alone is insufficient; a consistently wrong pair receives no joint credit. Missing-member coverage and presented candidate positions are explicit.

6 Software evidence and adversarial controls

The exact-oracle ceiling completes all 15 episodes with zero regret and scores all probes correctly. A privileged static rule computes the transitive closure of occupied blockers plus unwanted pieces, removes that set in legal order, then greedily installs missing target parts without prematurely blocking other missing parts. It also solves 15/15 episodes. This control selects no actions using the cost tables and performs no state-space search. The source graph remains private, so its success does not establish that the visual task is solved; it does refute a claim that these episodes inherently require deep search given perfect relations.

Depth-1, depth-2 and depth-4 bounded search use the private transition model, Hamming leaf estimates and visited-state tie-breaking, without cost oracles. All solve 9/15 tasks in each repeated run. This non-improvement is a property of these controls, not a model capability result. The public structured greedy policy uses only target flags, type compatibility, observed IDs and rejection memory; its three seeds solve 2, 2 and 3 episodes. Uniform legal-action random has a privileged validity set. Blind random samples public P/S IDs without reading pixels. Neither random policy is forced to select known traps. No control observes fabricated model outputs or hand-inserted wrong answers.

Semantic validation replays all 15 witnesses in both observation tracks, checks all 36 matched pairs in private and public form, and exercises self-induced diagnosis after successful actions, restoration, recurrence, censoring, invalid attempts, report census, missing survival, and replay tampering. HTTP checks cover both tracks, all probe kinds, revision and idempotence, persisted view budgets, source-route rejection and study restrictions. Six Firefox sessions complete injected repair and reconfiguration for every structure through actual UI controls, including inspection, diagnosis, replay, public export and feedback. These are software acceptance trials using reference actions.

7 Limitations and unresolved review concerns

This release repairs measurement and one demonstrated probe shortcut; it does not resolve the small number of source families, establish support rerouting, or show empirical model shortsightedness. Relations can be recovered or memorized in a public small bank, so held-out source acquisition and external replication remain necessary. No visual-perception or human study has been conducted. Explicit image tracks make such studies possible but do not provide their results. Image catalogs and fixed sites also simplify correspondence.

The physical abstraction excludes alternative insertion paths, rotation, simultaneous subassembly moves, gravitational dynamics, clutch forces, hand access and actual construction. Renderer fidelity is not a validation of these omitted phenomena. There is no claim of general D2–D4 difficulty, maximum planning horizon, CVPR readiness, or immunity to unknown shortcuts. A strong next release needs new independently qualified mechanisms that challenge the static relation baseline, more source groups, and preregistered image-versus-structured model and human experiments.

References

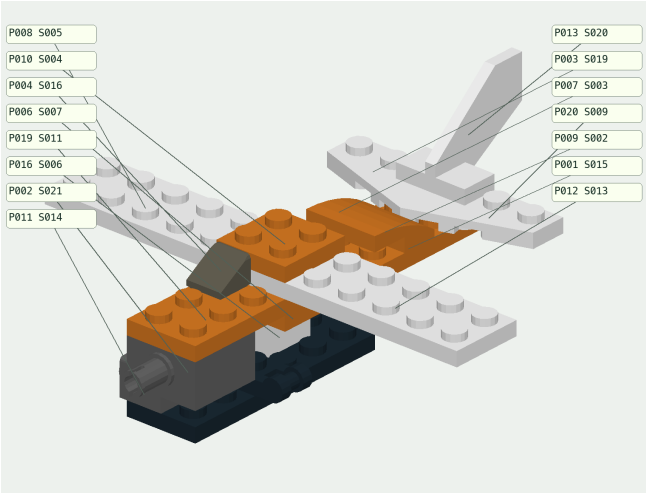
[1] LTRON / Break and Make. arXiv:2207.13738. https://arxiv.org/abs/2207.13738.

[2] InstructioNet. arXiv:2410.01111. https://arxiv.org/abs/2410.01111.

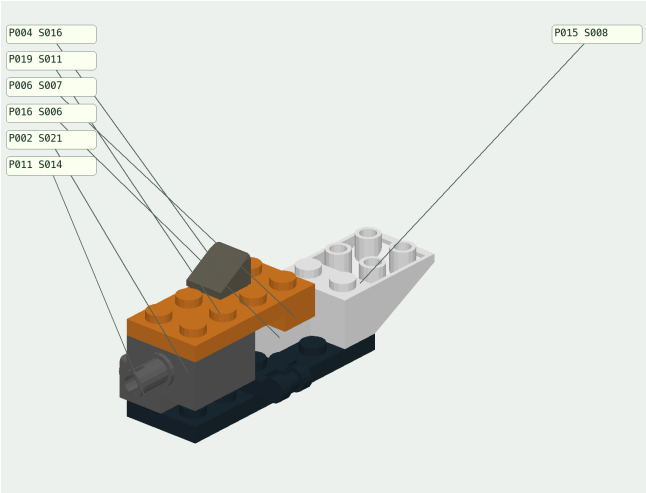
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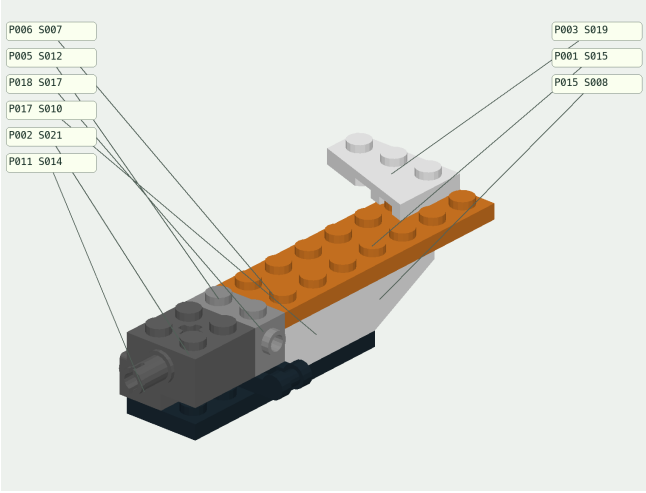
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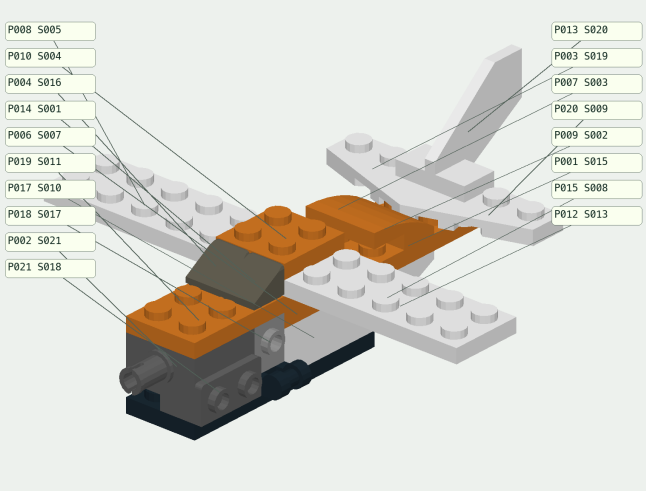
(a) Reconfiguration start, action 0.



(b) After 10 successful actions.



(c) After 20 successful actions.



(d) Exact partial target, action 34.

Figure 1: Authoritative airframe reconfiguration replay, identity seed 31, rendered from original LDraw triangles with source poses unchanged. Eight initially mismatched sites require 34 successful actions and at least 15 detachments under the certified fixed paths. All displayed states remain connected to the clamped base. Despite this cost, the privileged static relation rule solves the task; length and rework do not certify deep search. Frame hashes and exact occupancy are evaluator-side provenance artifacts.